

CONGRESSIONAL TWITTER ENGAGEMENT

Statistical Modeling, Temporal Patterns, and Strategic Insights for Lobbyists4America

Authors

Sahana Dasgupta

Affiliations

UCSD

Introduction

This study analyzes congressional Twitter activity for Lobbyists4America to optimize digital advocacy and public outreach. By evaluating 548 political profiles and 880,057 tweets, it investigates how media, posting timing, and follower counts drive retweets and likes. Ultimately, these data-driven insights help strategists maximize audience reach and engagement in competitive political environments.

Methodology & Workflow



Loaded 548 user profiles and 880,057 tweets into Pandas dataframes to examine baseline numerical features and missing values.



Applied a robust two-step parsing function to standardize mixed Unix epoch timestamps and string dates while identifying sparse geolocation columns.



Joined user metadata directly to tweet engagement records using a One-to-Many relational structure.



Ran statistical tests (T-Test, ANOVA, Spearman) to evaluate core hypotheses, extracted top political keywords using TF-IDF, and created custom engagement formulas.

Analysis & Textual Insights

TF-IDF Keyword Scores: rt (0.166), amp (0.117), today (0.067), great (0.045), house (0.033), new (0.031), thanks (0.029), work (0.022), support (0.022), and jobs (0.021)

Dominant Themes:

- Policy and Economy: High emphasis on legislative work (house, act, jobs, work, etc)
- Constituent Engagement: Community outreach and public acknowledgments (today, thanks, support, etc)

Datasets & Data Cleaning

Datasets: Merged users_df (N=548) and tweets_df (N=880,057)

Schema (ERD): One-to-Many entity relationship between User and Tweet, with recursive self-referencing for replies and quote tweets

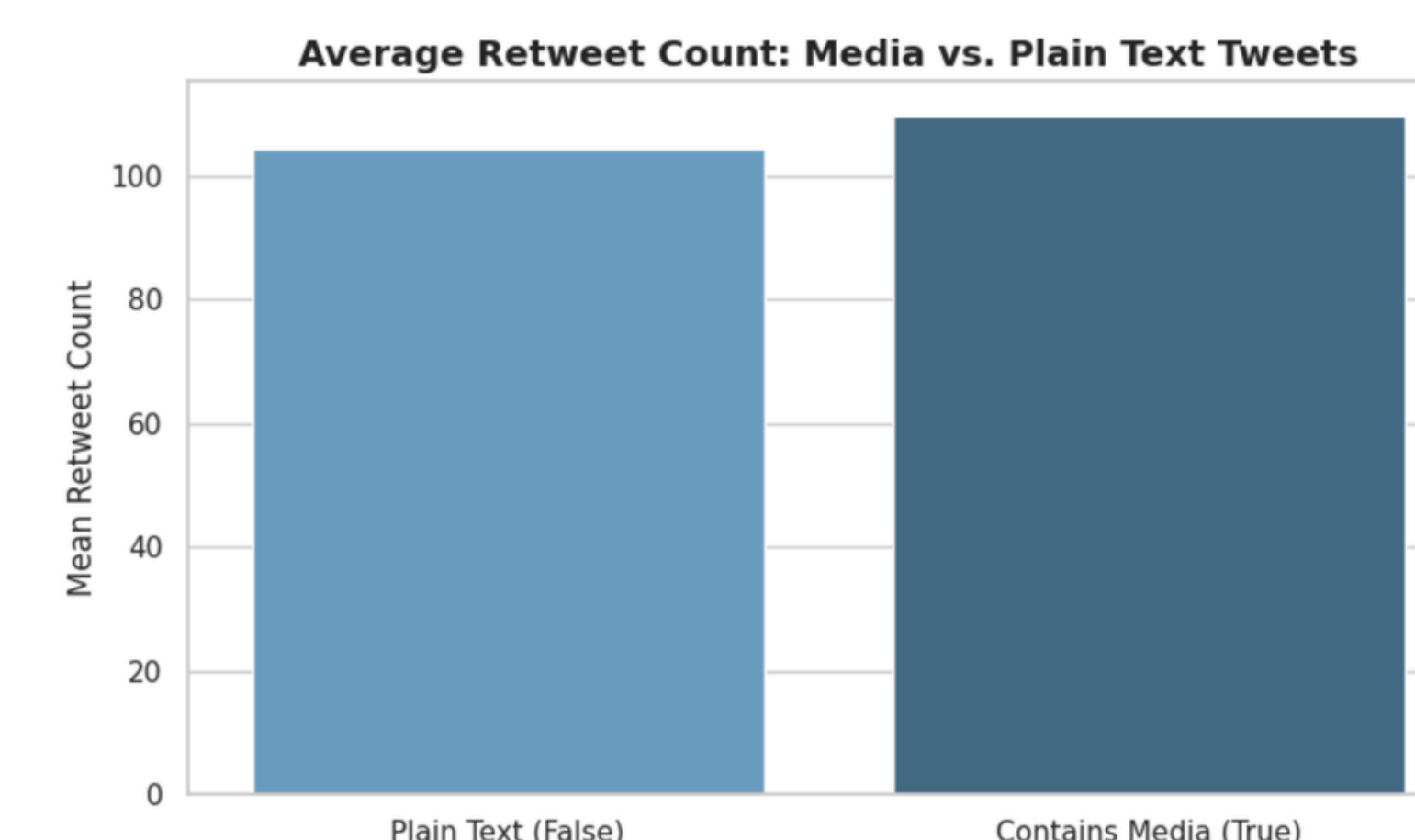
Data Preprocessing Steps:

- Two-step element-by-element parsing for mixed Unix epoch and string timestamps in the created_at column, representing when tweets were posted
- Handling extreme amounts of missing values in geolocation attributes (>98% nulls)
- Managing skewed distributions using non-parametric stats such as medians, IQR, and Spearman's rho

Hypothesis 1

Statistics: Welch's t = 0.2596, p = 0.7952, α = 0.05

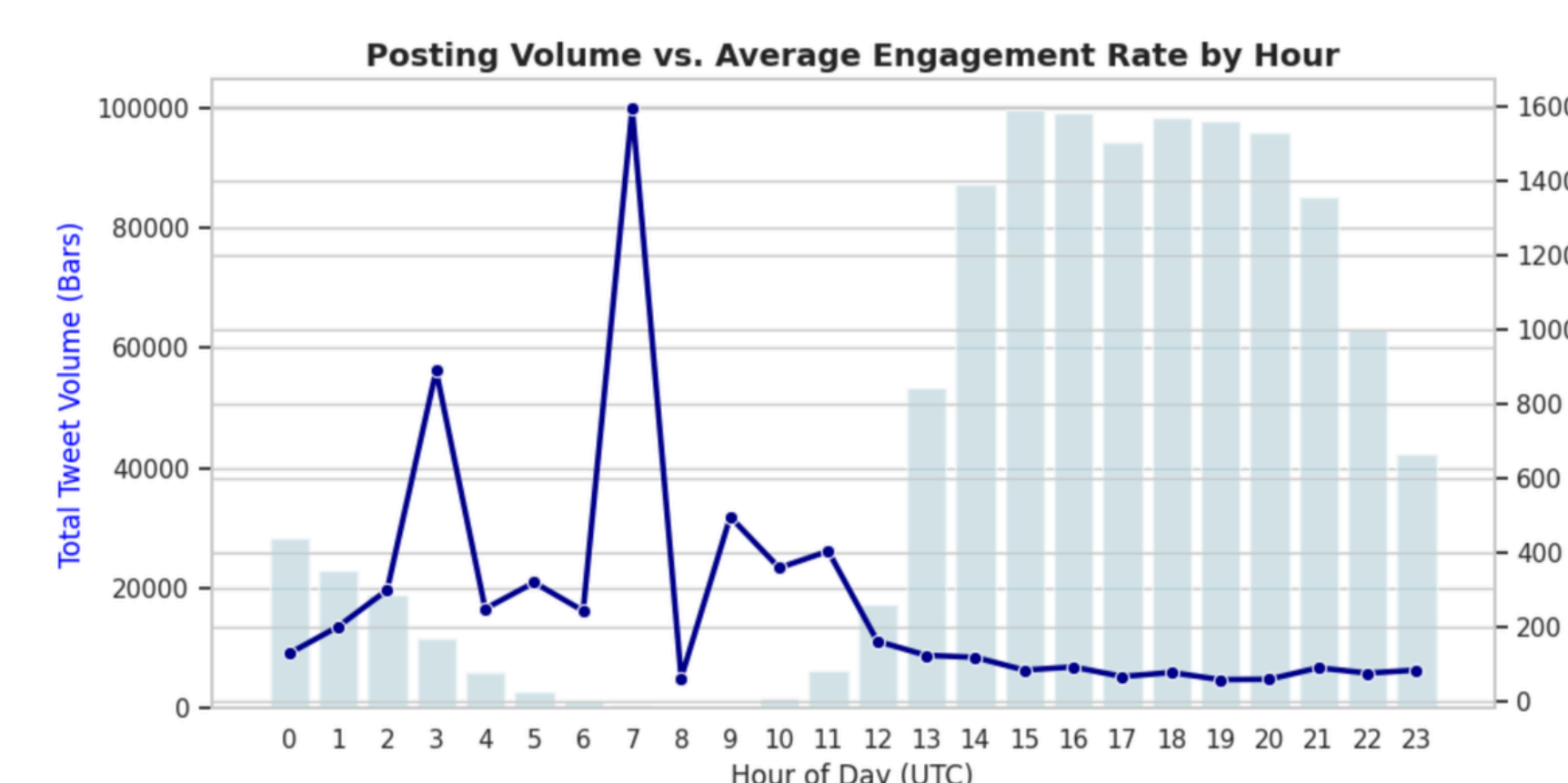
Findings: Sample means show higher average retweets for media (85.10 vs 59.21), but high variance renders the difference **statistically non-significant**.



Hypothesis 2

Statistics: ANOVA F = 15.3297, p = 9.6673×10^{-61} (**significant - disproven**)

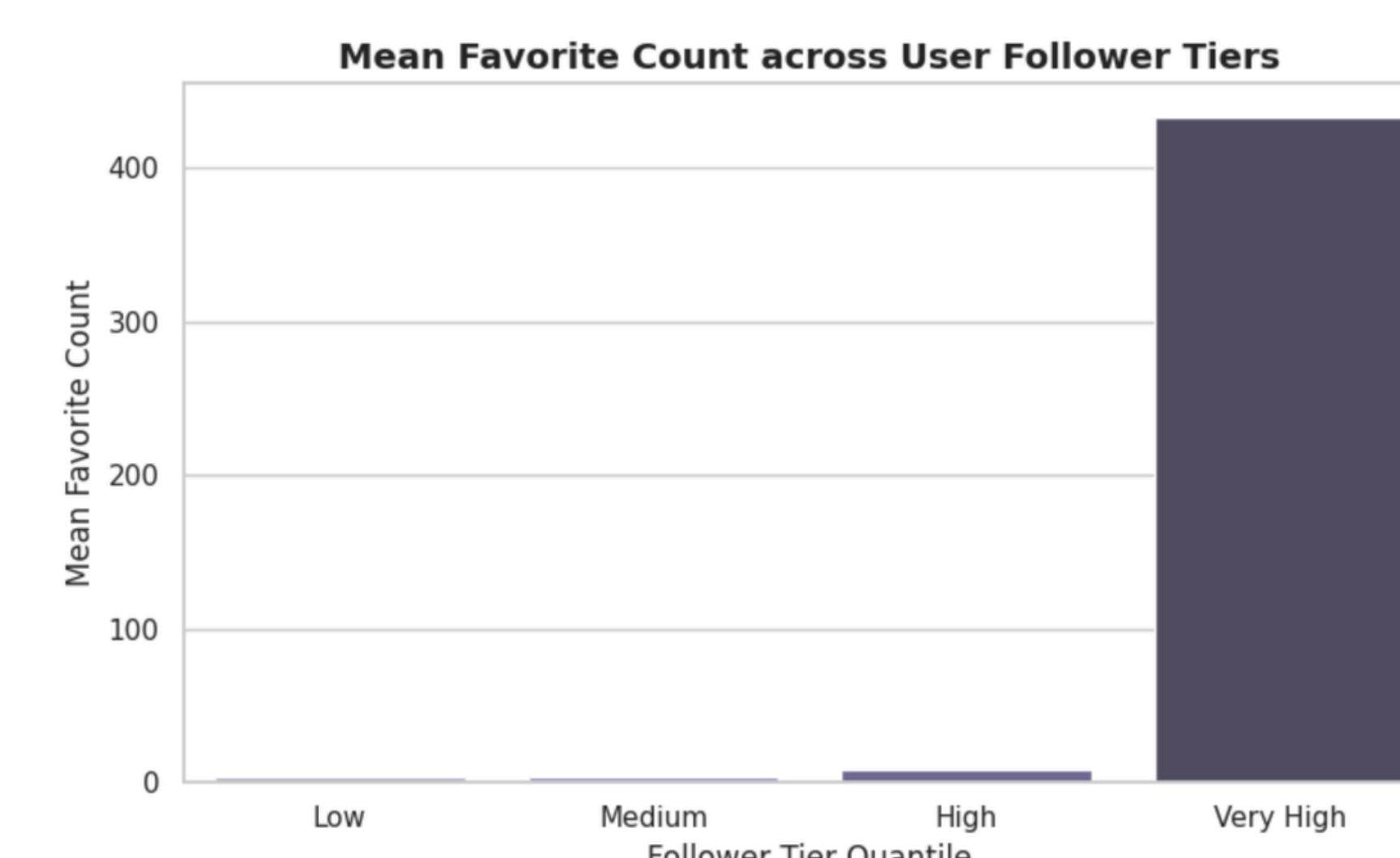
Findings: Peak posting volume (14:00–20:00 UTC) coincides with diluted tweet engagement due to feed saturation. Off-peak hours achieve peak interaction.



Hypothesis 3

Statistics: Spearman ρ = 0.2958, p < 0.0001, α = 0.05 (**significant - proven**)

Findings: Influencer scale operates on a non-linear tipping point, where only the top 25% "Very High" tier earns exponential likes (173.33).



Initial Hypotheses & Methodology

H1: Tweets with media (entities.media) earn significantly higher retweets/favorites

H2: Tweeting during peak platform volume yields higher per-tweet engagement

H3: Very high follower counts drive disproportionately high engagement

Methods: Welch's T-Test, One Way ANOVA, Spearman Correlation, Linear Regression, and TF-IDF Vectorization

Conclusion + Recommendations

Analysis Takeaways:

- **Optimize Schedule:** Shift high-priority announcements to lower-volume time windows (02:00 UTC - 07:00 UTC) to avoid feed competition
- **Target Top Quantiles:** Focus major legislative campaigns on verified, "Very High" tier accounts for viral amplification
- **Leverage Micro-Influencers:** Track engagement_per_follower to partner with highly response micro-accounts